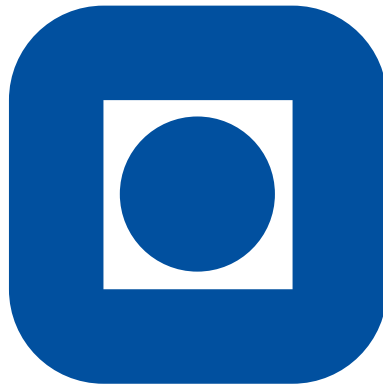
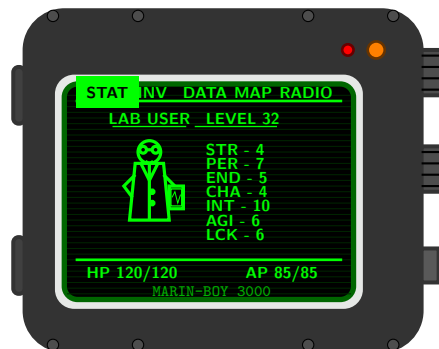

LABORATORY USER'S SURVIVAL GUIDE



NTNU



DON'T PANIC

If the model is upside down (in reality or only in the data),
the coordinate frames are backwards,
the data stream has vanished,
or the experiment is producing revolutionary and/or unexplained
results,

take a deep breath and keep reading.

Most problems begin with somebody pressing the wrong button.

Most solutions begin with somebody reading the manual.

You are likely one, or both, of these beings.

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1 Introduction

Welcome, Lab User. This guide serves as a practical reference for the systems and equipment used throughout the laboratories. In addition to describing the purpose and operation of each system, it contains troubleshooting procedures for the most common issues that may be encountered during laboratory work.

The objective of this guide is to enable users to diagnose and resolve simple problems

independently, particularly outside normal working hours when immediate assistance may not be available.

The guide is organized by system type. Each section introduces a particular system, while laboratory-specific configurations and procedures are presented in dedicated subsections where necessary

2 Motion Capture System

2.1 Startup Procedure

This section describes the standard startup procedure for most users of the motion capture system. If your project requires a different setup or configuration, contact the laboratory engineers before proceeding.

2.1.1 Quick Startup Checklist

2.1.2 Qualisys Track Manager Startup

Open QTM (Qualisys Track Manager). When the *Manage Projects* window appears, select `Qualisys_Default<NAME OF LAB>`. Press `Ctrl + N` to open a new measurement window.

Verify that the number of connected cameras displayed in the status bar matches the number of cameras currently in use.

In the toolbar on the left, select *3D View*. This displays the laboratory coordinate system, which is defined as follows:

- The x -axis points toward the wavemaker.
- The z -axis points downward.
- The y -axis completes the right-handed coordinate system and points from the gangway toward the opposite side of the basin.
- The origin is placed as close to the center of the basin as possible. At the time of writing, its exact position may vary between calibrations.
- The coordinate system orientation and origin location are established during calibration, which is normally performed by laboratory personnel. If your project requires the exact origin position or a different coordinate system orientation, contact the laboratory engineers before starting.

Press `Ctrl + D` to open the *Data Info* window. By default, this window displays marker data and coordinates.

Right-click anywhere in the window and select *Display 6DOF Data*. The window will then display the position and orientation of all defined rigid bodies:

x , y , z , roll, pitch, yaw

The displayed data also includes the *residual*¹.

Note that QTM displays position coordinates (x , y , and z) in millimeters and orientation angles (roll, pitch, and yaw) in degrees.

You may now place the vessel in the basin. If the vessel is tracked correctly in QTM and the residual remains sufficiently low, the system is ready for use. Keep in mind that residuals may increase in regions with limited camera coverage or areas that were not fully included during calibration. The calibrated volume can be visualized by pressing **Alt + 7**.

If the vessel is not detected correctly or the residual remains high (> 1 mm), continue to Section 2.1.3.

2.1.3 Rigid Body Setup and Verification

If the model has not already been defined, or if you are using a different vessel, the corresponding rigid body definitions must be loaded or created.

Open the *Project Options* window by clicking the gear icon in the upper-left toolbar. Navigate to *Processing* → *6DOF Tracking*.

If the model has not previously been defined, or if you need to create a new rigid body definition, follow the procedure described below.

Place the model in the basin. Only one model should be present during rigid body acquisition. Remove other models from the basin or cover their markers to prevent false detections. It is important that it is in view of as many cameras as possible (clicking *2D* on the left takes you back to *camera view* which is helpful here). The initial position (x, y, z) doesn't matter, but make sure that the vessel heading is aligned with the desired heading corresponding to 0° yaw, if it is important for your experiment also make sure that the vessel is balanced, the reason for this is that when you acquire a rigid body QTM sets $(roll, pitch, yaw) = (0, 0, 0)$.

Go to *6DOF settings* window in the *Project Options*, press *Acquire Body* button. Verify that the body has been detected and then click *Apply* and then *OK*. Close the measurement window by pressing the lower of the two crosses in the top right corner. Open a new measurement window and repeat the steps up until you see the rigid body coordinates in 6-DOF.

For the first time user it might seem that everything is in order. One should be able to see the rigid body defined by the markers as well as its coordinates in 6DOF, but QTM uses the centroid of the markers defining a rigid body as the body frame origin which does not correspond to the model geometric center. To remedy this all models should have a reference marker. Lab engineers should have provided you with its position relative to the model geometric center and the model waterline

Find the marker in QTM corresponding to the reference marker on the model. Select the marker and note its label. The label will typically have the format *Body1-x* or a similar name if the rigid body has previously been renamed. Return to *6DOF Tracking* settings, select the rigid body and click *Translate*.

¹The rigid body residual is the average error, measured in millimeters, between the observed marker positions and the corresponding points in the rigid body definition. https://docs.qualisys.com/qtm/content/qtm_user_interface/qtm_windows/6dof_data_information.htm?Highlight=residual

So that point [marker label] in the body has local coordinates (in mm):

Enter the values for the model in question and press *OK*, press apply and verify that the body has moved the origin of the system.

2.2 Common Failure Modes and Symptoms

2.2.1 Incorrect Body Frame Origin

Symptoms

- The model appears offset from its expected position in the global reference frame.
- Reported position does not correspond to the physical location of the model.
- Position appears shifted relative to other tracked objects.

Possible Causes

- The rigid body origin in QTM was defined at the centroid of the marker constellation instead of being translated to the desired model reference point.
- The reference marker coordinates were entered incorrectly.
- The rigid body definition was recreated or modified without applying the required translation.

Recommended Actions

- Verify the location of the rigid body origin in QTM.
- Use the model reference marker whose position relative to the model's geometric center and waterline is known.
- In the 6DOF tracking settings, use the *Translate* function and assign the reference marker its known coordinates (x, y, z) in the model reference frame.
- This translates the rigid body coordinate system such that the body frame origin coincides with the desired model reference point.

2.2.2 Low Marker Count

Symptoms

- Fewer markers than expected are visible in QTM.
- 6DOF tracking is unstable or unavailable.

Possible Causes

- Cameras are blocked by personnel or equipment.
- Markers are covered by water spray or reflections.
- One or more markers have detached from the model.
- Poor camera exposure settings.

Recommended Actions

- Verify that all markers are attached and clearly visible.
- Remove any obstructions from the camera field of view.
- Check the camera images for poor contrast or excessive reflections.

2.2.3 High Residuals

Symptoms

- Residual values exceed the normal operating range (≈ 1 mm).
- Position and orientation estimates appear noisy.
- Sudden jumps occur in the measured pose.

Possible Causes

- Markers have shifted due to a collision or other physical impact.
- Incorrect rigid body definition.
- Marker misidentification.
- Inadequate camera coverage.
- Calibration inaccuracies.

Recommended Actions

- Verify that the correct rigid body definition is loaded.
- Check that all markers are correctly identified.
- Move the model to an area with better camera coverage.
- Remove the rigid body from 6DOF tracking and reacquire it.

2.2.4 Tracking Dropouts

Symptoms

- The model disappears from the 6DOF data stream.
- Position data contains gaps.
- The body repeatedly loses and regains tracking.

Possible Causes

- Too few markers are visible to the camera system.
- Temporary obstruction by personnel or equipment.

Recommended Actions

- Check marker visibility in the 3D view.
- Ensure that at least three markers remain visible to the system.
- Remove obstacles from the capture volume.

2.2.5 Camera System Mismatch

Symptoms

- The number of connected cameras differs from the expected configuration.
- One or more camera views are missing.
- QTM displays camera connection warnings.

Possible Causes

- A camera is powered off.
- Network or synchronization cables are disconnected.
- An incorrect QTM project was loaded.
- A camera may be experiencing network or synchronization issues. In this case, contact laboratory personnel.

Recommended Actions

- Confirm that the correct project is loaded.
- Verify that all cameras are powered and connected.
- Contact laboratory personnel if a camera remains unavailable.

2.3 Screenshots and Examples

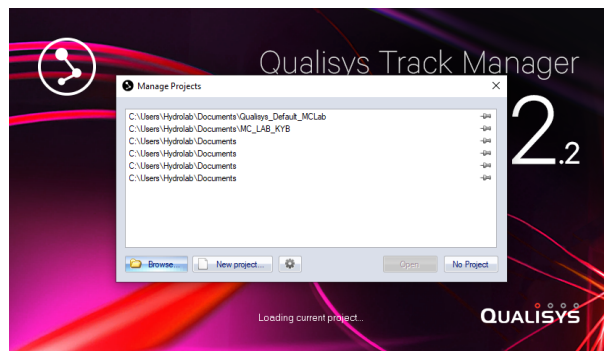


Figure 1: The first window that appears when starting QTM. Here you will select your project or create a new one



Figure 2: Start a new measurement window by clicking the white sheet in the top left or pressing CTRL + N. You must also repeat this if the qualisys system crashes.



Figure 3: Camera view in *marker mode* click on the *3D* icon on the left to enter *3D mode*.

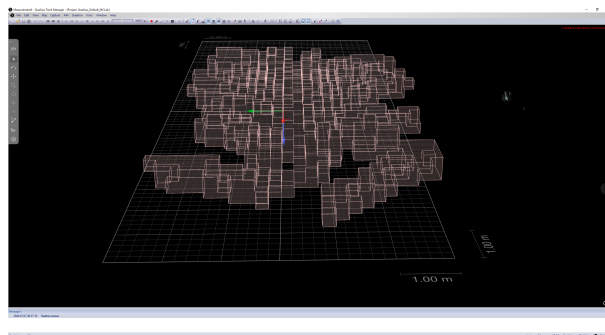


Figure 4: Lab frame of reference with the calibrated volume shown. Press CTRL + D or alternatively click *view* on the top toolbar and then select *Data Info 1* to show marker positions in *2D*.

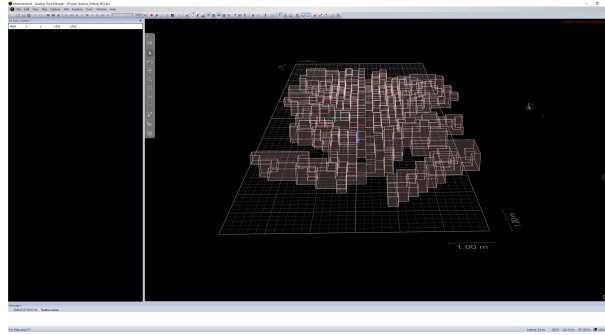


Figure 5: Lab frame of reference with the calibrated volume and 2D marker position shown. Right click and select *6DOF* to show rigid body coordinates.

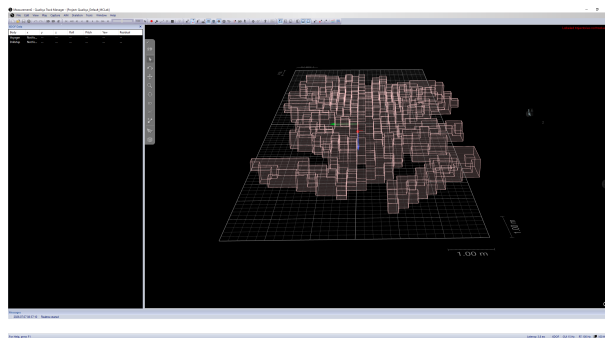


Figure 6: Lab frame of reference with the calibrated volume and rigid body coordinates.

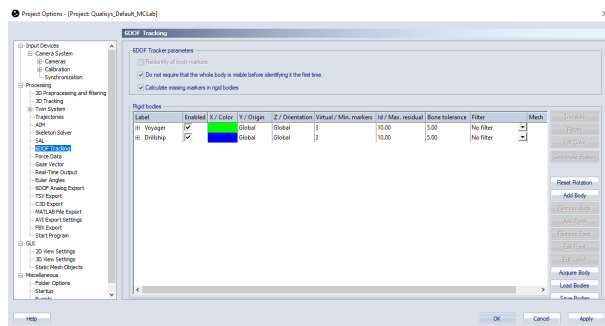


Figure 7: *6DOF Settings* window where you can *Add*, *Acquire*, *Load*, and *Save* rigid bodies.

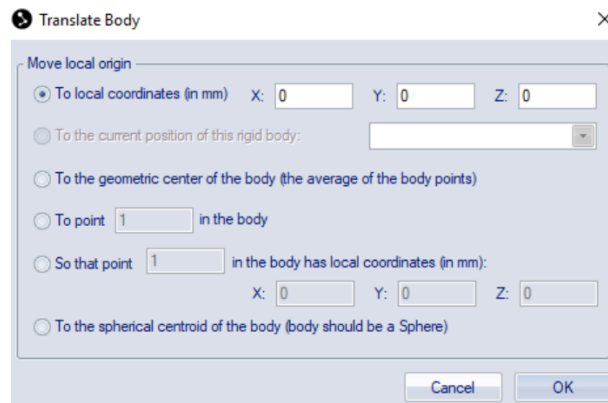


Figure 8: *Translate Body* Settings where we can move the local origin.